# Jashanpreet Singh

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*Amritsar, Punjab, India*

## Education

**Guru Nanak Dev University - MSc. FYIP Physics - CGPA 7.8/10** **2024 – Present**
*Amritsar, Punjab, India*

**The Millennium School - Non Medical - 84.2%** **2022 – 2024**
*Patiala, Punjab, India*

**The Kaintal School - Secondary Education - 94.4%** **2010 – 2022**
*Patiala, Punjab, India*

## Achievements

### Academic Grants & Research Infrastructure

- **IUCAA Pegasus Cluster Computational Resource Grant** for JWST LRD Research During my Undergrad. **(2026)**

### Academic Awards

- **ICNPA** (International Conference on Nuclear Physics and Application) **Best Oral Presentation Award**. **(2024)**
- **JEE Mains** (Joint Entrance Examination) **AIR 231**. **(2024)**
- **NSEA** (National Standard Examination For Astronomy) **Scholar**. **(2023)**
- **The Breakthrough Junior Challenge Finalist**. **(2021)**
- **IAAC** (International Astronomy and Astrophysics Competition) **Bronze Honours**. **(2021)**

### Other Awards and Honours

- **IAAC** (International Astronomy and Astrophysics Competition) **Ambassador**. **(2024 - Present)**
- **ISRO** (Indian Space and Research Organisation) **Space Tutor**. **(2023 - Present)**
- **APY** (Astronomy Photographer of the Year) **Shortlisted**. **(2021)**
- **SDNP** (South Down National Park Astrophotography Competition) **Runner Up**. **(2021)**

### Hackathons

- **AlgOlympia, Chandigarh University** (National Hackathon) **Winner**. - 50000 INR **(2026)**
- **Ideathon, GNDU** (Golden Jubliee Center of Entrepreneurship and Innovation) **Winner**. - 15000 INR **(2025)**
- **HACKOWASP7.0** (North India's Biggest Hackathon, TIET) **Runner Up**. - 150000 INR **(2025)**
- **Ad Astra** (National Astronomy Ideathon, NIT, Jalandhar) **Second Runner Up**. - 5000 INR **(2025)**

## Publications

**1. A Unified Deep Learning Framework for Automated Morphological Classification and Photometric Analysis using DECals Legacy Imaging** **(MNRAS Positive Review)**

- Galamo is an open-source Python package for comprehensive galaxy analysis, integrating machine learning and statistical methods. It provides automated tools for morphology classification, kinematics, photometry, and spectral analysis to aid astrophysical research.
- *Authors: Jashanpreet Singh Dingra,* ***Hum Chand*** *&* ***Madhu Sudan*** — www.galamo.org

**2. Investigating the relationship between black hole mass and galaxy dynamics in Seyfert Type II and LINER galaxies** **(MNRAS Under Review)**

- Presented, for the first time, the correlation between $M_{\rm BH}$ and $\sigma$ specifically for Seyfert Type II, an analysis made challenging due to obscuration by nuclear dust and the presence of a torus.
- *Authors: Jashanpreet Singh Dingra and Madhu Sudan*

## International Collaborations

**1. The Evolutionary Destiny and Observational Deficits of JWST "Little Red Dot" Analogues in IllustrisTNG** **(On Going - 2026)**

- By tracking their main descendant branches forward to $z = 0$, we predict these early-universe anomalies are the direct progenitors of today's massive Brightest Cluster Galaxies (BCGs). Furthermore, a stacked temporal analysis aligned to the quenching epoch definitively proves that radiatively efficient AGN feedback is responsible for their rapid star-formation cessation.
- *Authors: Jashanpreet Singh Dingra,* ***Akanksha Kapathia (Max Planck Institute of Astronomy, Heidelberg, Germany)***

2. **Exploring Active Galactic Nuclei in the Sloan Digital Sky Survey: A Multiwavelength Study** **(2026)**
   - Project conducted under the **Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences, Warsaw.**
   - *Mentors & Collaborators:* ***Marzena Sniegowska (Astronomical Institute of the Czech Academy of Sciences) and Biswaraj Palit (Copernicus Astronomical Center of the Polish Academy of Sciences)***

## Recent Research Projects

1. **Pipeline to distangle AGN flux contribution from total galaxy flux using PFVG Method** **(2025 - Present)**
   - This project focusing on the study of the AGN flux variability using PFVG method from COMOS Casata Servey.
   - *Authors: Madhu Sudan, Jashanpreet Singh Dingra, Ankit Sarkar & Hum Chand*

2. **Galspar & RAGA - Astronomy Pipelines for Galactic Morphological Study via GALFIT** **(2025 - Present)**
   - Developed 'galspar', its primary purpose is to perform precise photometric and morphological measurements on astronomical images of galaxies stored in the standard FITS format.
   - Developed 'RAGA', for automative pipeline for statistically morphology study via GALFIT.
   - *Authors: Jashanpreet Singh Dingra, Hum Chand and Madhu Sudan (Central University of Himachal Pradesh)*

3. **Measuring Stellar Velocity Dispersion of SDSS Galaxies using a MILES Template via our Fitting Pipeline: vdisp_fit** **(2025 - Present)**
   - **IUCAA Visiting Student Project**
   - Developed 'vdisp_fit', a Python pipeline to measure stellar velocity dispersion in galaxies using SDSS data.
   - The method involves direct spectral fitting by convolving a MILES stellar template with a Gaussian broadening function and utilizes a robust $\chi^2$ minimization to derive kinematic parameters.
   - *Authors: Jashanpreet Singh Dingra and Gulab Chand Dewangan (IUCAA)*

4. **Deep Learning Model for Alzheimer Detection and MRI Brain Segmentation** **(2025)**
   - Engineered a convolutional neural network (CNN) to classify Alzheimer's disease stages from brain MRI scans.
   - The model also performs semantic segmentation to identify and delineate key brain structures affected by the disease, such as the hippocampus, to aid in early diagnosis.
   - *Authors: Jashanpreet Singh Dingra and Hardeep Kaur*

5. **Study of Stellar Dynamics of Open Star Clusters Using the Runge-Kutta Method** **(2025)**
   - Implemented an N-body simulation using the fourth-order Runge-Kutta method to model the gravitational interactions and long-term dynamical evolution of stars within open clusters.
   - Analyzed stellar trajectories and cluster stability over cosmic timescales to understand dissolution processes.
   - *Authors: Jashanpreet Singh Dingra and Suprit Singh (IIT Delhi)*

6. **Crab Pulsar Spin Down Rate Over 5 Years (2015–2023)** **(2024)**
   - Analyzed archival X-ray timing data of the Crab Pulsar from 2015 to 2023 to precisely measure its spin-down rate and braking index.
   - The study confirmed the expected rotational energy loss due to magnetic dipole radiation and investigated short-term timing noise and glitches.
   - *Authors: Jashanpreet Singh Dingra, Pratham Jain, Uttakarshika and Manam Tiwari*

## Major Computational Astronomy Contributions

**Astropy - Astrophysics and astronomy python package** **(2025 - Present)**
- *Contributed in astropy.coordinates and many more modules.*

## Conferences, Schools & Talks

**IISER Mohali Introductory School on Astronomy and Astrophysics**, IISER Mohali **(2026)**
**ESA Workshop: Machine Learning meets Galaxy Classification 2**, ESAC, Spain **(2026)**
**Himalayan School for Training in Astronomy Research (HiSTAR)**, UPES (Dehradun) by IUCAA **(2026)**
**School on Galactic Astrophysics using UVIT, HST and JWST**, IUCAA, Muvattupuzha, Kerala **(2026)**
**4th BINA Conference**, Kerala, India **(2025)**
**AdventureX, China's Biggest Hackathon**, Final Round, Hangzhou, China **(2025)**
**Conference on Advances in Cosmology**, Christ University, Bangalore **(2025)**
**Summer School on Analysis and Statistical Modelling of Space Science Data**, MCNS, Karnataka **(2025)**
**ZTF Summer School 2025**, University of Minnesota, USA **(2025)**
**13th IIST Astronomy and Astrophysics School (IAAS)**, Kerala **(2025)**

**Summer School on Gravitational-Wave Astronomy**, ICTS-TIFR, Bangalore **(2025)**
**National Conference on Active Galactic Nuclei**, CUHP, India **(2025)**
**Invited Speaker @ Star Party**, The Millennium School, Patiala, Punjab **(2025)**
**International Conference on Nuclear Physics and Its Applications**, New Delhi **(2024)**
**68th Symposium on Nuclear Physics**, IIT Roorkee **(2024)**
**Speaker at Astronomy Webinar**, Astronomy Club, Kosovo **(2023)**
**Pulsar Data Analysis**, HEASARC (NASA), India **(2022)**

## Investigations

**SFR Correlation with the Mass and Metallicity of Galaxies from SDSS DR7** **(2023)**
- Conducted an analysis revealing a positive correlation between the star formation rate (SFR) and the stellar mass of galaxies, indicating that more massive galaxies tend to have higher star formation rates.

**Correlation Between the Recessional Velocity and Distance of Type Ia Supernovae** **(2022)**
- Identified a strong positive correlation between the distance of a Type Ia supernova and its recessional velocity, consistent with Hubble's Law. Notably discovered a supernova with a recessional velocity of $0.8c$, located approximately 1000–1500 Mpc away.

**Simulation that Predicts Mars Oppositions** **(2020)**
- Developed a Python-based simulation to predict Mars oppositions over the next 100 years, aiding in observational planning and planetary alignment studies.

**24-Hour Sun Analysis from SDO Data** **(2020)**
- Analyzed solar activity using 24-hour observational data from the Solar Dynamics Observatory (SDO) on December 26, 2020, focusing on sunspot activity and solar flares.

## Funded Startups

**Sitee** (sitee.in) **2025 – Present**
*Founder & CTO*
- Founded and developed Sitee, an AI-powered platform that generates full-stack websites from natural-language prompts using a custom 1.2B+ parameter LLM.
- Integrated vector databases, including FAISS and Pinecone, for context retrieval, code generation, and scalable AI workflows.

**Brepline** (brepline.vercel.app) **2025 – Present**
*Founder & CTO (Early Stage)*
- Developing a 7.9B+ parameter large multimodal model (LMM) for AI-driven 3D modeling, generating parametric CAD models in STEP/B-Rep formats from natural-language prompts and sketches.
- Integrating AI with SolidWorks through custom plugins for AI-assisted CAD design, parameter editing, and automated engineering workflows.

## Other International Publications

**Shortlisted in International Astronomy Photographer of the Year Award** **(2021)**
- Published in: **Forbes**, **BBC**, **Royal Museums Greenwich**

**South Down National Park International Release** **(2021)**

## Leadership / Social Work

**Recognized by the Municipal Commissioner for heroically saving a child during a bus accident.**
**Member of the Art of Living's Punjab Flood Relief Team.**
**Tech Head of Guru Nanak Dev University Physics Club**
**Founder of Dingrastro Club, a global astronomy community with over 200 members.**
**Active Member of Over 7 Astronomy Clubs Worldwide**

## Technical Skills

**Programming Languages**: Python, C++, C, R, FORTAN, MERN, Assembly, MATLAB, Bash

**Database Systems**: PostgreSQL, MySQL, MongoDB, Redis, GraphQL, Elasticsearch

**Cloud & DevOps**: AWS, GCP, Azure, Docker, Kubernetes, Airflow, Git, Linux

**Tech Stack**: FastAPI, LangChain, LangGraph, REST API, Weights & Biases, MLflow

**Libraries / Frameworks**: PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace, OpenCV, PIL

**Machine Learning**: Transformers, GNNs, State Space Models (Mamba), RAG, LLM Fine-tuning (SFT, LoRA, QLoRA, PEFT), Agentic AI

**Quant Tools**: Statsmodels, cvxpy, JAX, TA-Lib, QuantLib, PyPortfolioOpt

**Visualization**: Matplotlib, Seaborn, Plotly, Power BI, Tableau

**Core CS Fundamentals**: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA)